

Translations of tangent functions



- 1 Which of the following is the correct domain of $y = \tan x$?
 - A All real numbers, except integer multiples of 90° .
 - B All real numbers, except odd integer multiples of 90° .
 - C All real numbers, except even integer multiples of 90° .
 - D All real numbers.

- 2 How has the graph $y = \tan x + 4$ been transformed from $y = \tan x$?

- 3 State whether the following statement is true or false:
The graph of $y = \tan\left(x + \frac{\pi}{4}\right)$ has a horizontal translation to the left from $y = \tan x$.

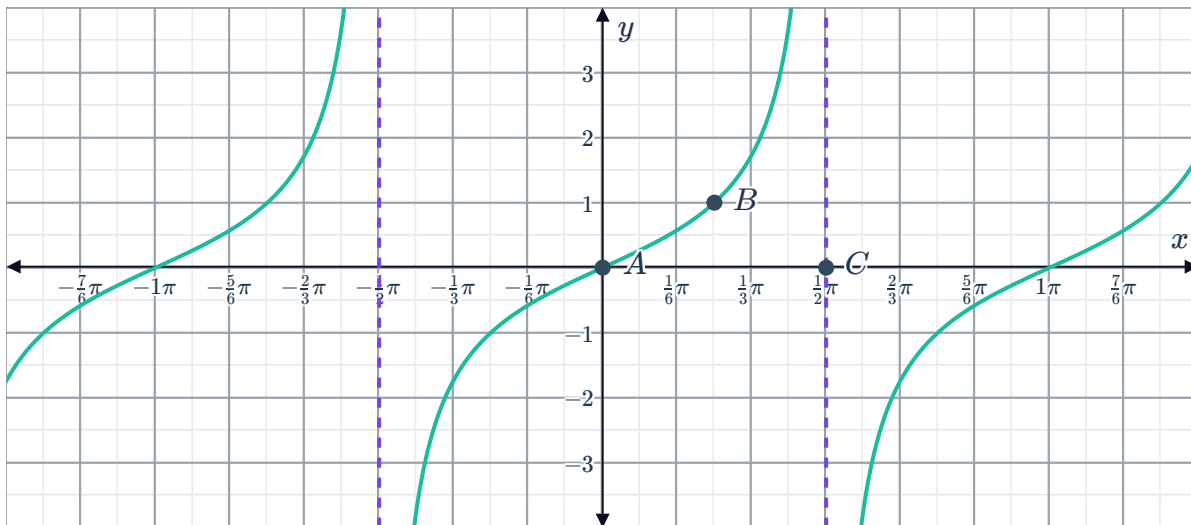
- 4 Consider the function $y = \tan(x - 90^\circ)$.
 - a Does the function have y -intercept?
 - b Determine the period of the function.
 - c How far apart are the asymptotes of the function?
 - d State the first asymptote of the function for $x > 0$.
 - e State the first asymptote of the function for $x \leq 0$.
 - f Sketch the graph of the function.

- 5 Consider the function $y = \tan(x - 60^\circ)$.
 - a Determine the value of y at the y -intercept.
 - b Determine the period of the function.
 - c How far apart are the asymptotes of the function?
 - d State the first asymptote of the function for $x \geq 0$.
 - e State the first asymptote of the function for $x \leq 0$.
 - f Sketch the graph the function.

6 The function $f(x) = \tan\left(x - \frac{\pi}{6}\right)$ is to be graphed on the interval $\left[\frac{2\pi}{3}, \frac{8\pi}{3}\right]$.

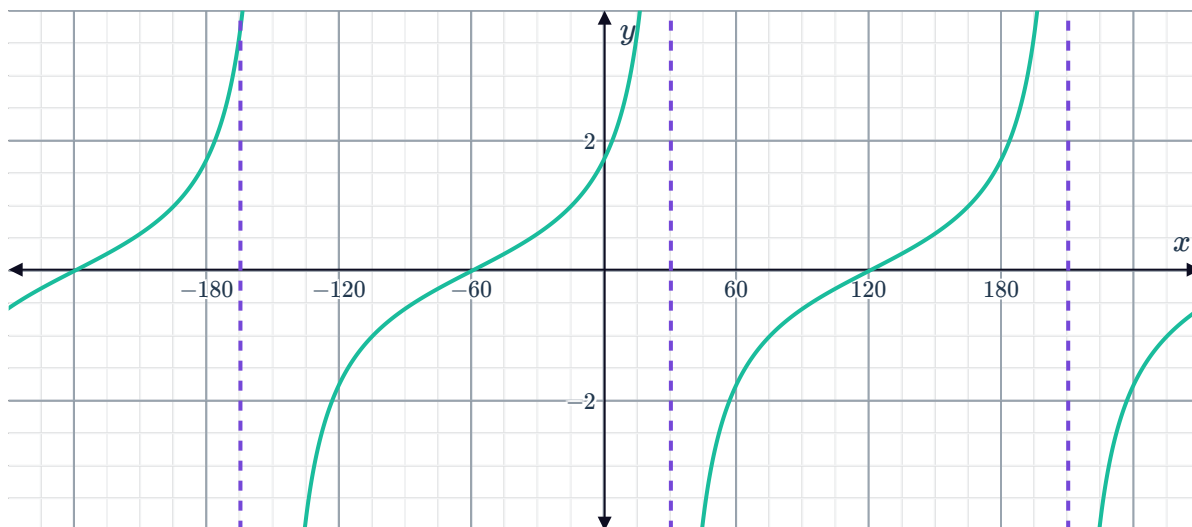
- a Find the equations of the asymptotes of the function that occur on this interval.
- b Find the x -intercepts of the function that occur on this interval.
- c Sketch the function $f(x)$ on the interval.

7 Consider the graph of $f(x) = \tan x$ and three points $A(0,0)$, $B\left(\frac{\pi}{4}, 1\right)$ and $C\left(\frac{\pi}{2}, 0\right)$:



- a If $f(x)$ undergoes a transformation to $g(x) = \tan\left(x - \frac{\pi}{3}\right)$, state the coordinates of the following points after the transformation:
 - i A
 - ii B
 - iii C
- b Describe the transformation of $f(x)$ to $g(x)$.
- c Now, sketch the graph of $g(x)$ on the domain $-\pi \leq x \leq \pi$.

8 Consider the graph shown:



a Determine whether the following could be the equation of the graph:

i $p(x) = \tan(x - 120^\circ)$

ii $q(x) = \tan(x + 60^\circ)$

iii $r(x) = \tan(x - 1020^\circ)$

iv $s(x) = \tan(x - 680^\circ)$

b Now, which of the following statements is true?

A The graph of $y = \tan(x - h)$ is the same as the graph of $y = \tan(x - h - 180n)$, for any integer n .

B The graph of $y = \tan(x - h)$ can be reflected in the y -axis to make the graph of $y = \tan(x - h - 180n)$, for any integer n .

C The graph of $y = \tan(x - h)$ is different for every value of h .

D The graph of $y = \tan(x - h)$ is the same as the graph of $y = \tan(x - h - 180n)$ only when $n = \pm 1$.

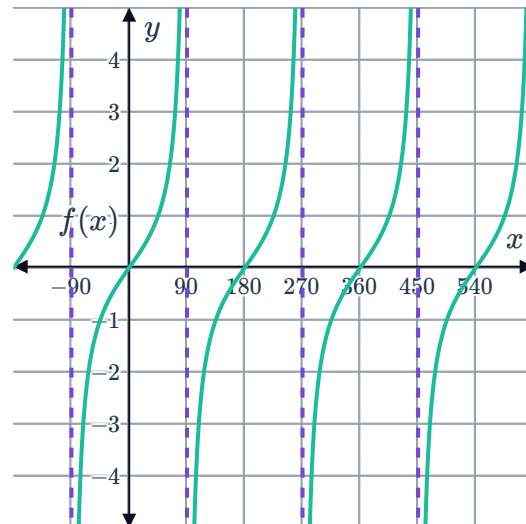
9 Let $f(x) = \tan x$ and $g(x) = \tan(x - 45^\circ)$.

What can be said of the domain and range of both functions? Explain your answer.



Dilations of tangent functions

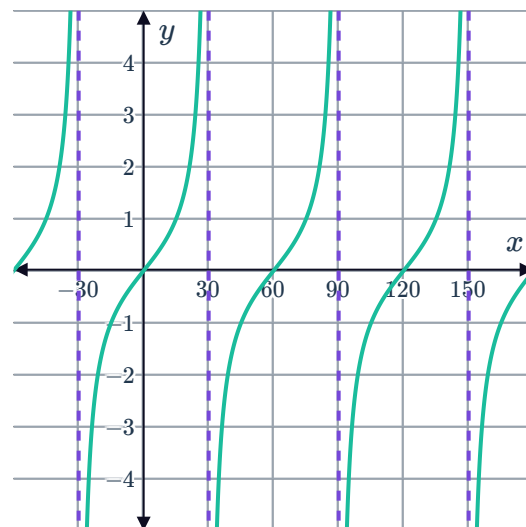
- 10 The graph of $f(x) = \tan x$ is shown.
On the same coordinate plane, sketch the graph of $g(x) = \tan\left(\frac{1}{2}x\right)$.



- 11 Consider functions of the form $y = \tan bx$.
- a Complete the table, identifying the period of the function when $b = 1, 2, 3, 4$.
 - b State the period of $y = \tan bx$.
 - c As the value of b increases, describe the effect on the period of $y = \tan bx$.

Function	Period
$y = \tan x$	π
$y = \tan 2x$	
$y = \tan 3x$	
$y = \tan 4x$	

- 12 Consider the graph of a function in the form $f(x) = \tan bx$:
- a State the period of the function.
 - b Now, state the equation of the function.



13 An asymptote of a function of the form $g(x) = \tan bx$ is $x = \frac{\pi}{8}$. Find the equation of $g(x)$.

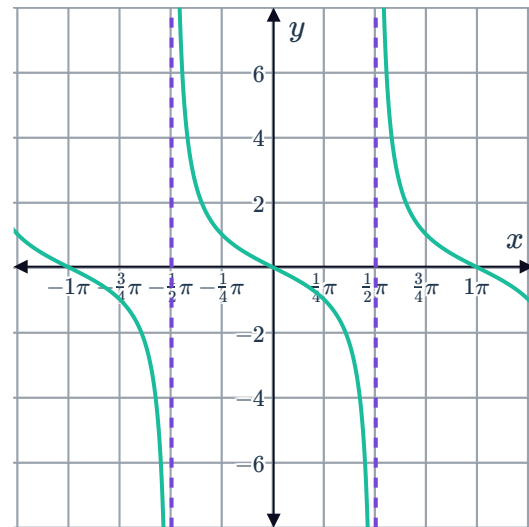
14 How has the graph $y = 3 \tan x$ been transformed from $y = \tan x$?

15 On the same coordinate plane, sketch the graphs of $y = \tan x$ and $y = \frac{1}{2} \tan x$, on the domain $-2\pi \leq x \leq 2\pi$.

16 Describe how the graph of $y = -5 \tan x$ has been transformed from the function $y = \tan x$.

17 Consider the graph of the given tangent function:

- The given graph is the result of what transformation from $y = \tan x$?
- State the equation of the graphed function.



18 For each of the following functions:

- Determine the y -intercept.
- Determine the interval between the vertical asymptotes of the function.
- Now, state the period of the function.
- State the equation of the first asymptote of the function for $x \geq 0$.
- State the equation of the first asymptote of the function for $x \leq 0$.
- Sketch the graph the function on the domain $-\pi \leq x \leq \pi$.

a $y = -\tan x$

b $y = \tan \left(x + \frac{\pi}{3} \right)$

c $y = \tan \left(\frac{x}{2} \right)$



Series of transformations

19 For each of the following functions:

- i** Find the value of y when $x = \frac{\pi}{4}$.
- ii** Determine the period of the function.
- iii** Now, state the interval between the asymptotes of the function.
- iv** State the equation of the first asymptote of the function for $x \geq 0$.
- v** State the equation of the first asymptote of the function for $x \leq 0$.
- vi** Sketch the graph the function on the domain $-\pi \leq x \leq \pi$.

a $y = 5 \tan x + 3$

b $y = 4 \tan 3x$

20 Consider the function $y = \tan \left(2x + \frac{\pi}{3} \right)$:

- a** Rewrite the function so that it is in the form $y = \tan (a(x + b))$.
- b** Now, sketch the graph of $y = \tan \left(2 \left(x + \frac{\pi}{6} \right) \right)$.

21 Sketch the graph of the following functions:

a $y = \tan (2(x - 45))$

b $y = \tan (2x + 90^\circ)$

22 Consider the graphs of the following functions and state whether or not the graph is the same as $y = -\tan x$:

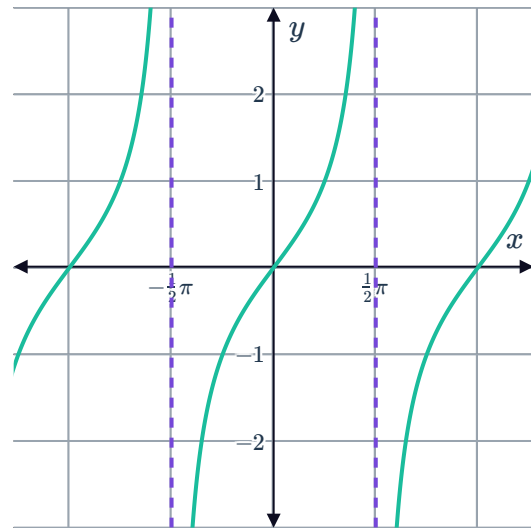
a $y = -\tan \left(x + \frac{3\pi}{4} \right)$

b $y = -\tan (x + \pi)$

c $y = -\tan \left(x + \frac{\pi}{2} \right)$

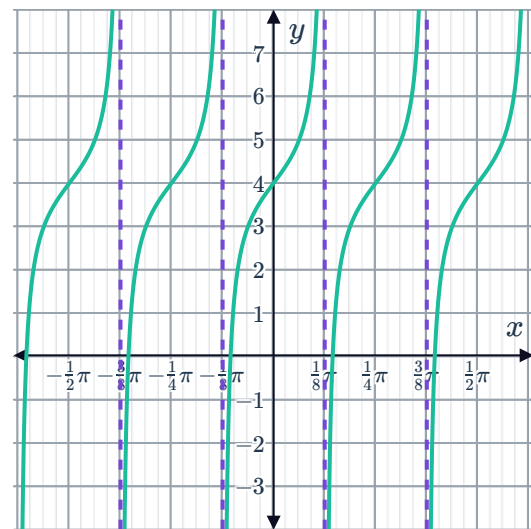
d $y = -\tan (x + 2\pi)$

- 23** The graph $y = \tan x$, as shown, is reflected over the y -axis and then translated vertically 3 units down. Find the equation of the new graph.

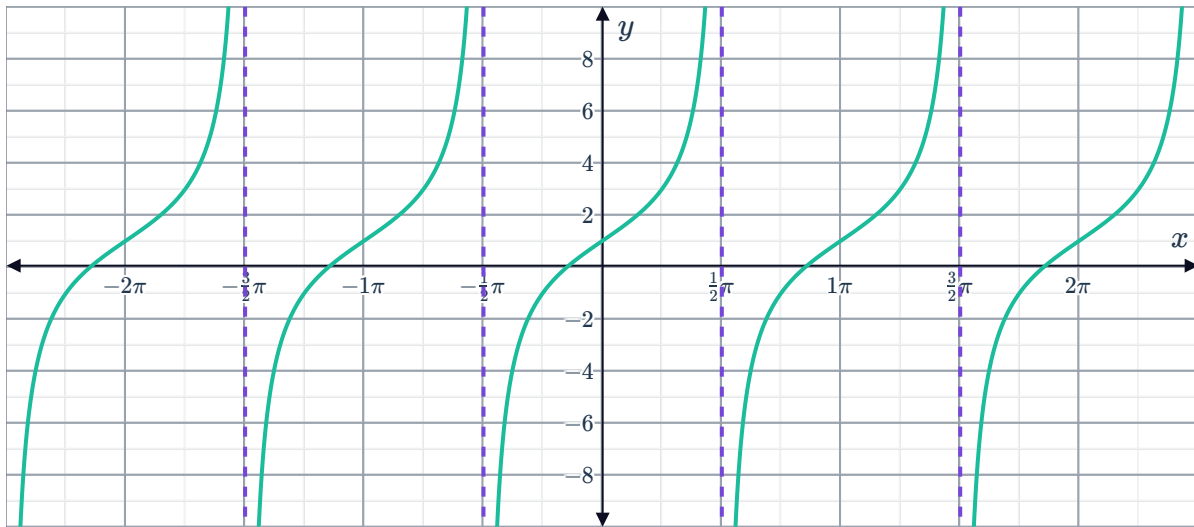


- 24** Consider the graph of the given tangent function:

- The given graph is the result of what transformations on $y = \tan x$?
- State the equation of the graphed function.



25 Consider the given graph of $f(x) = 2 \tan x$



a State the domain of $f(x)$.

b State the range of $f(x)$.